

American Mathematical Society

Response to the RFI: AI Action Plan

Networking and Information Technology Research and Development
(NITRD) National Coordination Office (NCO), National Science Foundation.

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The American Mathematical Society (AMS) is the largest professional organization for mathematicians in the world. Founded in 1888 to further the interests of mathematical research and scholarship, it serves the national and international community through its publications, meetings, advocacy and other programs, which

- promote mathematical research, its communication and uses,
- encourage and promote the transmission of mathematical understanding and skills,
- support mathematical education at all levels,
- advance the status of the profession of mathematics, encouraging and facilitating full participation of all individuals,
- foster an awareness and appreciation of mathematics and its connections to other disciplines and everyday life.

AMS is one of the world's leading publishers of mathematical books and journals and operates the MathSciNet indexing service for mathematical sciences publications. It sponsors scientific meetings across the United States and around the world, administers awards and grant programs, and promotes public awareness of mathematics.

Protecting intellectual property (IP) in training large language models (LLMs)

IP, copyright, and attribution are problematic areas in the training of LLMs. LLMs essentially hoover up all available data, and thus will include openly available published content (book and journal), retracted content, preprints (arXiv for example), pirated content and so on. The question of copyright is important in an AI framework, and there are discussions, for example in the UK, on changing copyright laws to accommodate AI technologies:

<https://www.theguardian.com/technology/2025/feb/11/uk-copyright-law-consultation-fixed-favour-ai-firms-peer-says>

On the one hand LLMs do not provide attribution as content is broken into relatable elements, not assigned to an author. This is important, especially in math, as the ideas expressed in a journal article or book are the IP – generally not being a report on an experiment, which is more common in the empirical sciences. There are also research integrity concerns, as the loss of attribution means a loss of accountability. Research output can only be as good as the quality of input. In addition, it is not clear that plausible looking output is correct output. LLMs frequently produce reasonable sounding statements that are factually false, as the algorithms do not incorporate logical checks and verifications of information.

Building on the existing mathematical sciences education infrastructure (K-16) to help citizens understand AI and critically evaluate the products of generative AI

The US education system struggles to train people in computer science and AI in particular. Even leading research universities struggle to recruit faculty versed in the latest techniques, as they compete directly with technology companies for talent. Secondary schools rely on existing teachers to cover new advances. Often, people educated in mathematics incorporate computing into their lessons. These teachers need support and professional development to teach the latest ideas. But even teachers in the humanities need to help students develop critical thinking skills, to evaluate whether AI-produced content is accurate or misleading.

Strategic investments in fundamental research in the mathematical and computational sciences that support key technologies make AI possible; and also research on how AI will transform research in existing fields (AIMing etc.)

Fundamental research in neural networks, machine learning, optimization, and graph theory made possible recent breakthroughs in AI. Continued leadership in these fundamental areas will seed future achievements. Federal support is essential for work that cannot be immediately incorporated into marketable products.

The state of AI research today can be likened to, in the words of Turing award winner Yann LeCun, building steam engines without knowing thermodynamics. Mathematics research is critically needed to develop a firm theoretical foundation for AI. This is necessary to create AI that is efficient, capable of correct and explainable decision making, and robust.

At the same time, AI has transformed long-standing research fields like optimization, scientific computing, fluid dynamics, and mathematical modeling. Even theoretical topics like number theory and knot theory have been transformed through these technologies. Future progress rests on support for databases and structured computations that can serve as training data for models. Again, federal support for this basic research will be crucial.

Active support for secure platforms for data analysis using rigorous frameworks like differential privacy

To maintain public confidence, data used for AI models must be curated to avoid situations where individuals can be identified in training data. Even “anonymized” datasets often fail to protect confidentiality. New paradigms like encrypted search - which allows queries without full access to a dataset - and differential privacy - should be pursued.

Investment in open source development and documentation to allow researchers to analyze performance

Open source code and documentation is essential for evaluating the relative effectiveness of various computational approaches to problem-solving. LLM-based solution techniques can yield dramatic results but their ultimate utility depends on the scale of up-front investments in training the models. Assessing the security, consistency, and fairness of models is difficult without access to the source code. Having publicly available code templates is essential for training the next generation of STEM workers.

Reasonable guidelines on international collaboration and export controls to allow fundamental research to continue unhindered

Current export-control regulations support international collaboration on fundamental research intended for publication in the scientific literature. The openness of American universities to students and researchers from around the world are key to their unique creativity and productivity. While proprietary technologies in commercial products and defense applications are essential to national security, a robust open research community helps the United States to recruit the best and brightest from around the world in basic science and mathematics. Many of these individuals remain in the United States, developing products and technologies essential for national security.

Frameworks for audits of automated decision processes to allow organization leadership and policy-makers to understand the aggregate impacts of the systems in use

Many processes in business and government are routinely audited to ensure compliance with applicable laws and regulations. Too often, computer automation is used to insulate decisions from evaluation or scrutiny. Mathematical modeling can be a powerful tool in assessing the policy implications of AI decision processes. But this is only possible with access to data on inputs and outputs of the algorithms.

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